

K941 — (1) FLAGSHIP FINAL GLANCE = PASS (fixes correct + better than asked → external-ready, pending Casey’s GO); (2) RATIFY Elie’s glueball self-catch (false precision against a moving lattice target — the ratios downgrade, the mass gap + integer ladder survive); (3) RATIFY Grace’s one-operator SPINE (the entire strong sector = the spectrum of one heat semigroup on one domain). A clean, honest close: the flagship is at the door, and the one over-precision that could have gone near Cal was caught in the team’s own toy first.

Keeper | 2026-07-26 Sun | Three ratifications. The flagship passes the final glance; Elie’s fish-detector caught false precision in his own glueball numbers (score σ , not dev%); Grace compressed the strong sector to one operator’s spectrum.

(1) FLAGSHIP FINAL GLANCE — PASS (external-ready)

Lyra applied both staged fixes, and better than the K-notes asked: - Line 114 (K935): now separates (i) identification ($\beta_0=11N_c-2N_f=21>0$, coefficients imported — “not a derivation of the number”) from (ii) geometric derivation of the SIGN (the $\tau \rightarrow 0$ spectral flow of $\exp(-\tau H_B)$ on $H^2(D_{IV}^5)$, one operator, no loop diagram). Cites F705 (siblings, not one operator); states the keystone “does not solve the (B) mass-gap”; “11/3 imported (consistency), the sign geometrically derived.” Crisp and honest. - §3 confinement (K937): scoped to (A) no free colored asymptotic states (Schur, *including the adjoint gluon*), explicitly NOT (B) area-law/mass-gap — the YM Millennium problem, not claimed; even handles the gluon-(A)-confined-but-(B)-screened subtlety. §126/§129 also scoped to (A). - .md/.pdf in sync (07-26 13:06), clean rebuild. - VERDICT: PASS. The flagship is EXTERNAL-READY. My final glance is the gate; going external is Casey’s GO (never push without approval). All five cold-read fixes landed.

(2) RATIFY Elie’s glueball self-catch — FALSE PRECISION against a moving target

Elie’s fish-detector fired on his OWN toy (4873/glueball), in exactly the direction the discipline demands. Ratified: - The catch: the catalog holds glueball ratios in TWO BST forms — linear-ladder (7/5, 3/2, matched to Morningstar–Peardon 1999) and newer (23/16, 17/11, matched to 2024 lattice). The two forms CROSS OVER when the lattice shifts ~3–4%. Web-confirmed: MP-1999 ($0^{++}=1730 \pm 130$, $2^{++}=2400 \pm 145$, $0^{-+}=2590 \pm 170$) vs 2024 ($0^{++}=1.5\text{--}1.7$ GeV, $2^{++}=2.2\text{--}2.4$) — the target MOVED and carries 50–170 MeV (~few–10%) uncertainty. - The principle (sharpened “score σ not dev%”): the glueball ratio’s σ is set by the LATTICE uncertainty (few %). A BST form matching to 0.007% is agreeing to $\sim 0.001\sigma$ against a target that could (and did) move — that is false precision, not a win. And when TWO different BST forms both fit inside the few-% bar and swap under a percent-level data wiggle, neither is forced — they are percent-level fits to a loosely-pinned number, not derivations. New standing discipline for the observable-ratio class: cap the claimed precision at the target’s uncertainty; a sub-% match to a few-%-pinned number is $\sim 0\sigma$ and cannot discriminate forms. - SURVIVES (banked): the mass gap $0^{++} = n_C$ (structural, the keystone’s IR end); the spectral ladder $\frac{1}{2}C_2 = \{3, 5, 6, 7\}$ = the four dynamical primaries (an integer identity, D-tier exact — the over-determination is real, not precision-matched). - DOWNGRADED: the specific glueball mass-map ratios (2^{++} , 0^{-+} , 1^{+-}) → STRUCTURAL / few-%, form-ambiguous — agreements at the lattice’s own σ , NOT sub-percent derivations. - CREDIT: Elie’s linear conformal-energy mechanism is a genuine win — the quadratic $m^2 \propto \text{Casimir}$ reading MISSED the tensor glueball; the linear energy (the $SO(2)$ dilatation eigenvalue on $H^2(D_{IV}^5)$) gives the tower from two BST-natural numbers ($\lambda_0=n_C$ ground; $\text{twist}=n_C/2$ for parity — the same n_C -odd half-integer as the lepton π -column). Casey’s “remember linear algebra” was the mechanism again, not just the method. The mechanism is banked; only the precision of the ratio-matches was over-stated. - The Tier-1 AF sign is UNTOUCHED — this is purely glueball-mass over-precision, not the sign.

(3) CATALOG RECONCILIATION (Grace + Keeper) — the over-precise entries

- ★ GRACE — re-tier the glueball ratio entries (T186, T1444, T2490 and `data/bst_constants.json` `r_glueball_01`) from sub-percent DERIVED to STRUCTURAL / few-% (lattice- σ -capped, form-ambiguous). They cannot all be “derived” at their quoted precisions against a moving lattice target. Keeper ratifies the tier. The mass gap $0^{++}=n_C$ and the $\frac{1}{2}C_2=\{3,5,6,7\}$ ladder keep their tiers.

(4) RATIFY Grace’s one-operator SPINE — the strong sector compressed

Ratified, and it is the right structure: the entire strong sector is the spectrum + spectral-flow of one operator — the color-sector Tier-0 heat semigroup $\exp(-\tau H_B)$ on $H^2(D_{IV}^5)$ (T2428) — read at different proper times: $-\tau \rightarrow 0$ (a_2) $\rightarrow$ AF sign (T2526); $\tau \rightarrow \infty$ (Shilov) $\rightarrow$ (A)-confinement (T2523); spectral gap $\rightarrow$ mass gap (T1271); eigenvalue tower $\rightarrow$ glueball spectrum (T1403); τ -flow $\rightarrow$ the running; the scale $\rightarrow \Lambda_{QCD}$ (T2167). - The graph now forks T2428 $\rightarrow$ all strong-sector observables as spectral readings, replacing the scatter of pairwise edges. This is why the morning’s pieces kept collapsing into each other (“one heat kernel, two ends”): they were never separate results, only different windows on one operator. Casey’s one-domain steer WAS the deep structure, not just hygiene — the same lesson as the flavor sector (the clean statement is the mechanism).

Net

Flagship external-ready (Casey’s GO). Strong sector = one operator’s spectrum, honestly tiered: AF-sign derived, (A)-confinement derived, keystone structural, the mass gap $0^{++}=n_C$ + the $\frac{1}{2}C_2$ ladder banked, the glueball RATIOS honestly downgraded to structural (false precision caught in-house). Nothing false on the books.

— Keeper K941, 2026-07-26. (1) Flagship final glance PASS (line-114 (i)/(ii) split + §3 (A)/(B) scope, both better than asked; .md/.pdf synced) $\rightarrow$ external-ready pending Casey’s GO. (2) Ratify Elie’s glueball self-catch: FALSE PRECISION vs a moving lattice target (web-confirmed MP-1999 $\rightarrow$ 2024 shift, $\sim$ few-10% σ); two BST forms cross under 3-4% $\rightarrow$ neither forced; new discipline: cap precision at target σ (sub-% match to few-%-pinned = $\sim 0\sigma$, can’t discriminate). Survives: $0^{++}=n_C$ (structural), $\frac{1}{2}C_2=\{3,5,6,7\}$ ladder (integer D-tier). Downgraded: glueball ratios ($2^{++}/0^{-+}/1^{+-}$) $\rightarrow$ structural/few-%. Credit: Elie’s linear conformal-energy mechanism (fixes what quadratic broke). Catalog: Grace re-tiers T186/T1444/T2490/`r_glueball_01` $\rightarrow$ structural. (3) Ratify Grace’s one-operator spine: strong sector = spectrum of $\exp(-\tau H_B)$ on $H^2(D_{IV}^5)$ (T2428 forks to all observables). See `[[Keeper_K936_RATIFY_AF_sign_FULL_PASS]]`, `[[Keeper_K938_Cal_88_RATIFIES_strong_sector_close]]`, `[[feedback_score_sigma_not_devpct]]`.